

Research Document on NOVA Buildathon 2026

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Attention Lapse Detection

COG-BCI Dataset

The **COG-BCI database** [1] is a multi-session, multi-task EEG dataset designed for research on **passive brain-computer interfaces** (pBCI). Each subject performed the same battery of cognitive tasks across three sessions while **63 EEG channels** were recorded at **500 Hz** with a **common-average reference**. For every condition, the dataset also provides trial-level **behavioral logs**, which makes it possible to align neural activity with overt performance.

Dataset Structure

The local copy follows a **BIDS-like layout**, organized per subject (sub-XX) and session (ses-S1–ses-S3):

- eeg/ – raw EEGLAB files (.set header + .fdt binary data) for each condition;
- behavioral/ – trial-level behavioral logs (.mat) mirroring the same conditions;
- chanlocs/ – channel names and 3D coordinates.

Each session contains the same task battery: the N-back task (0/1/2-back), the Flanker task, the Multi-Attribute Task Battery (MATB), the Psychomotor Vigilance Task (PVT), and resting-state blocks (eyes open / eyes closed) at the start and end of the session.

Structure of Data on Psychomotor Vigilance Task

The **Psychomotor Vigilance Task** (PVT) is a classic **sustained-attention** paradigm. Participants watch a screen and press a button as soon as a stimulus appears. Stimuli are presented at **pseudo-random inter-stimulus intervals** of 2–10 s, so the task cannot be solved by rhythm or anticipation; performance therefore tracks the current level of vigilance.

The primary outcome is the **reaction time** (RT) on each trial. Slow responses and **lapses** are the hallmark of reduced vigilance, which is observed under sleepiness, fatigue, and **mind wandering**. When the mind wanders, attention is diverted away from the stimulus, so the participant reacts late or not at all – precisely the lapse signature that the PVT measures. PVT performance is therefore a standard **behavioral proxy** for mind-wandering episodes, and, combined with the simultaneously recorded EEG, it allows searching for the neural correlates of attentional drift.

The PVT EEG is stored in **EEGLAB format**: a .set header file together with a .fdt binary file that holds the raw samples. Inspecting the MATLAB struct of PVT.set gives the following picture:

```
EEG =  
  filename: 'PVT.set'   datfile: 'PVT.fdt'  
  nbchan:   63          trials: 1  
  pnts:     306060      srate:  500  
  xmin:     0           xmax:   612.118  
  data:     [63×306060 single]  
  ref:      'common'  
  event:    [272×1 struct]
```

The various fields have the following meaning:

- nbchan: 63, srate: 500 – the recording contains **63 EEG channels** sampled at **500 Hz**.
- trials: 1, pnts: 306060, xmax: 612.118 – a **single continuous recording** of 306060 sample points, i.e. roughly **612 seconds (~10.2 minutes)**, that is **not** segmented into epochs.
- data: [63×306060 single] – the raw amplitude values (in microvolts, single-precision floats) laid out as **channels × time points**.

- ref: 'common' – the signal has already been **average-referenced**.
- chanlocs: [1×63 struct] – the channel locations, matching chanlocs/get_chanlocs.txt.
- event: [272×1 struct] – event markers for the 90 PVT trials (stimulus onsets and responses) together with session start/end markers; these events make it possible to **segment the continuous signal into trials**.
- datfile: 'PVT.fdt' – the companion binary file containing the actual samples.

Corrupted Behavioral Log

The behavioral log PVT.mat describes the same 90 trials at the performance level. In theory, taken together, PVT.set / PVT.fdt provide the continuous neural signal while PVT.mat provides the per-trial behavioral ground truth. However, an inspection on the PVT.mat files using **MD5** shows that the files seem to be **identical across sessions and subjects**, which is unexpected.

A further inspection of the EEG files shows that sub-17 and sub-27, sub-25 and sub-28 have identical EEG recordings for the PVT. Those four subjects are removed from the analysis, leaving 25 subjects with valid data.

Since we are training for a model that detects lapses from EEG, we need to have the reaction times for each trial in order to label the trials as **lapse** or **non-lapse**. Hence, we need to **reconstruct the behavioral log** from the EEG event markers.

We use the mne package to read the EEG files as raw and extract the event markers via raw.annotations. The event labeled as 13 indicates the **stimulus onset** and the event labeled as 14 indicates the **response**. The difference between the two timestamps gives the **reaction time** for each trial.

Training EEGNet Using the COG-BCI Dataset

In order to obtain a clean

References

- [1] M. F. Hinss, E. S. Jahanpour, B. Somon, L. Pluchon, F. Dehais, and R. N. Roy, *COG-BCI database: A multi-session and multi-task EEG cognitive dataset for passive brain-computer interfaces*. (Jul. 2022). Zenodo. doi: 10.5281/zenodo.6874129.